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Equipment : PAF Unit 3

KKS : LJ30HLA10AV010-001

Technology : Vibrasi, IR Thermography dan Tribology

Tanggal : September 2020

Status : Kuning

Finding : Vibrasi PAF unit 3 Bearing 1,3&4 Warning

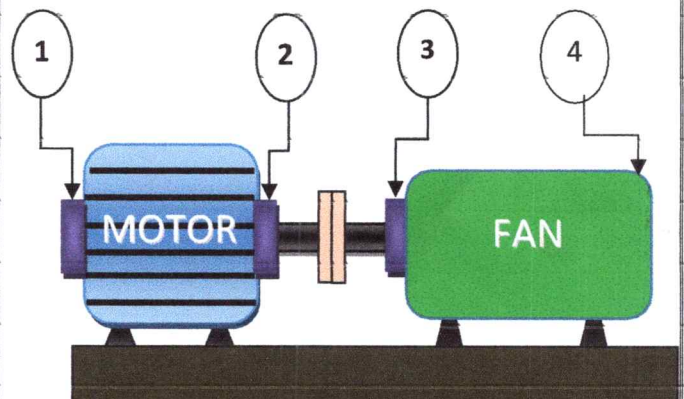
I. Data Spesifikasi Motor

MOTOR			FAN		
Type, Mfg	:	YKK 4504-4	Type, Mfg.	:	G6-30 19.50
Speed	:	1490	Speed	:	1485 r/min
Power	:	630	Power	:	630 Kw
Bearing Type	:	Roller Bearing	Flow Rate	:	400871 m ² /h
Inboard Bearing	:	NU226E/C3	Bearing Type	:	Journal Bearing
Outboard Bearing	:	6228/C3	Inboard Bearing	:	-
Rotor Bar	:	-	Onboard Bearing	:	-
Foundation	:	Rigid	Total Blade	:	-

FOTO EQUIPMENT



POINT PENGUKURAN



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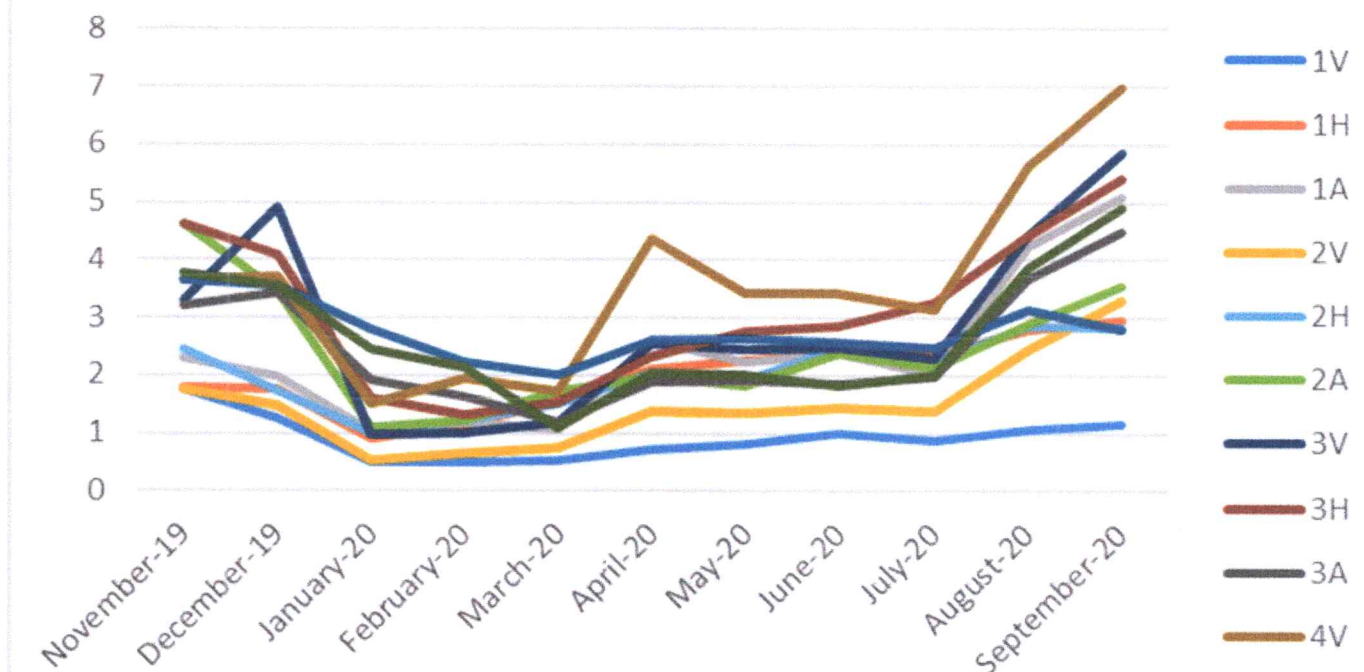
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II. Hasil Pengukuran Vibrasi

BULAN	BEBAN UNIT	ARUS (A)	DAMPER OPEN (%)	PAF UNIT 3												KETERANGAN
				MOTOR						FAN						
				1V	1H	1A	2V	2H	2A	3V	3H	3A	4V	4H	4A	
November-19	25.6	35.18	4.76	1.76	1.78	2.3	1.76	2.45	4.63	3.31	4.64	3.22	3.64	3.66	3.79	
December-19	25.6	35.18	4.76	1.25	1.77	1.98	1.48	1.74	3.47	4.92	4.09	3.43	3.71	3.54	3.55	
January-20	23.5	34.3	4.72	0.51	0.92	1.02	0.52	1.01	1.11	0.98	1.61	1.92	1.51	2.81	2.45	After ganti bearing Fan (3&4)
February-20	13	34.69	4.73	0.51	1.14	1.19	0.67	1.21	1.21	1.01	1.32	1.63	1.96	2.24	2.14	
March-20	25	35	3.71	0.53	1.58	1.07	0.76	1.52	1.72	1.2	1.53	1.18	1.72	2	1.11	
April-20	25.6	35	3.4	0.73	2.12	2.62	1.38	2.04	1.99	2.55	2.34	1.89	4.38	2.61	2.03	
May-20	25.6	67	3	0.82	2.26	2.22	1.34	1.91	1.83	2.46	2.77	1.92	3.43	2.64	2.01	
June-20	25.5	34.5	3.2	1.01	2.4	2.4	1.45	2.47	2.38	2.48	2.86	1.84	3.44	2.59	1.83	
July-20	25	34.8	3.3	0.88	2.37	2.02	1.37	2.26	2.13	2.28	3.26	1.99	3.13	2.47	2.02	
August-20	25.5	35	3.5	1.06	2.81	4.24	2.45	2.87	2.89	4.43	4.43	3.69	5.62	3.15	3.86	
September-20	25.2	35	4.35	1.15	2.96	5.09	3.3	2.83	3.56	5.84	5.42	4.49	6.98	2.79	4.9	

TRENDING VIBRASI PAF UNIT 3



Hasil Rekaman Spectrum Vibrasi :

✓ Pada Motor :

- Pada Bearing 1 (DE Motor) terjadi vibrasi Alarm dengan nilai 5.09 mm/s RMS, Indikasi Kerusakan Bearing

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✓ **Pada Pompa :**

- Nilai Vibrasi tertinggi di bearing 4 dengan nilai 6.96 status Warning, indikasi Kerusakan bearing

Rekomendasi :

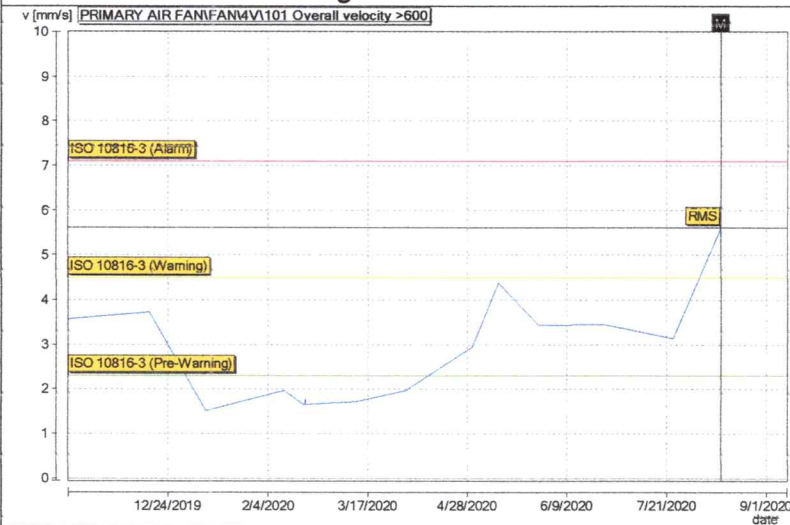
✓ **Pada Motor :**

- Penggantian bearing 1 dan 2

✓ **Pada Pompa :**

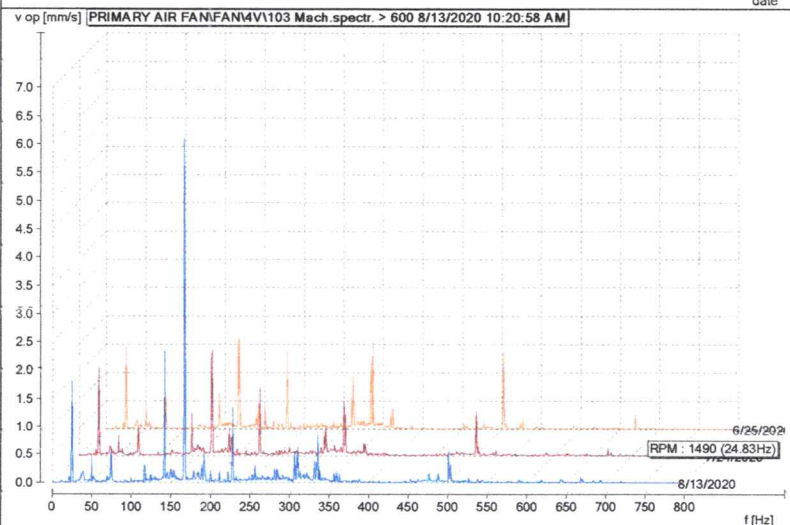
- Penggantian bearing 3 dan 4

Pengukuran Vibrasi



Keterangan

Hasil pengukuran vibrasi secara overall tertinggi pada bearing 4 arah vertikal 6.96 mm/s RMS, status prewarning. Berdasarkan ISO 10816-3 group1 (alarm 7.1)



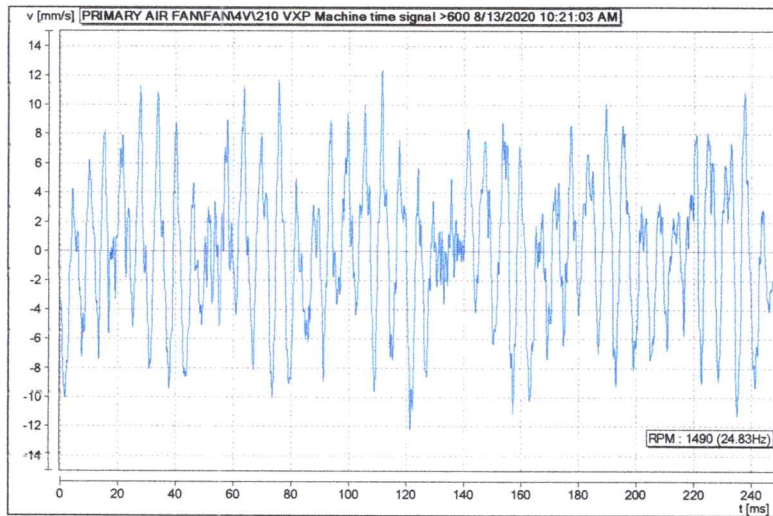
Bentuk syntom hasil pengukuran pada bearing 4 arah horisontal

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Indikasi kerusakan bearing dari pola time waveform,

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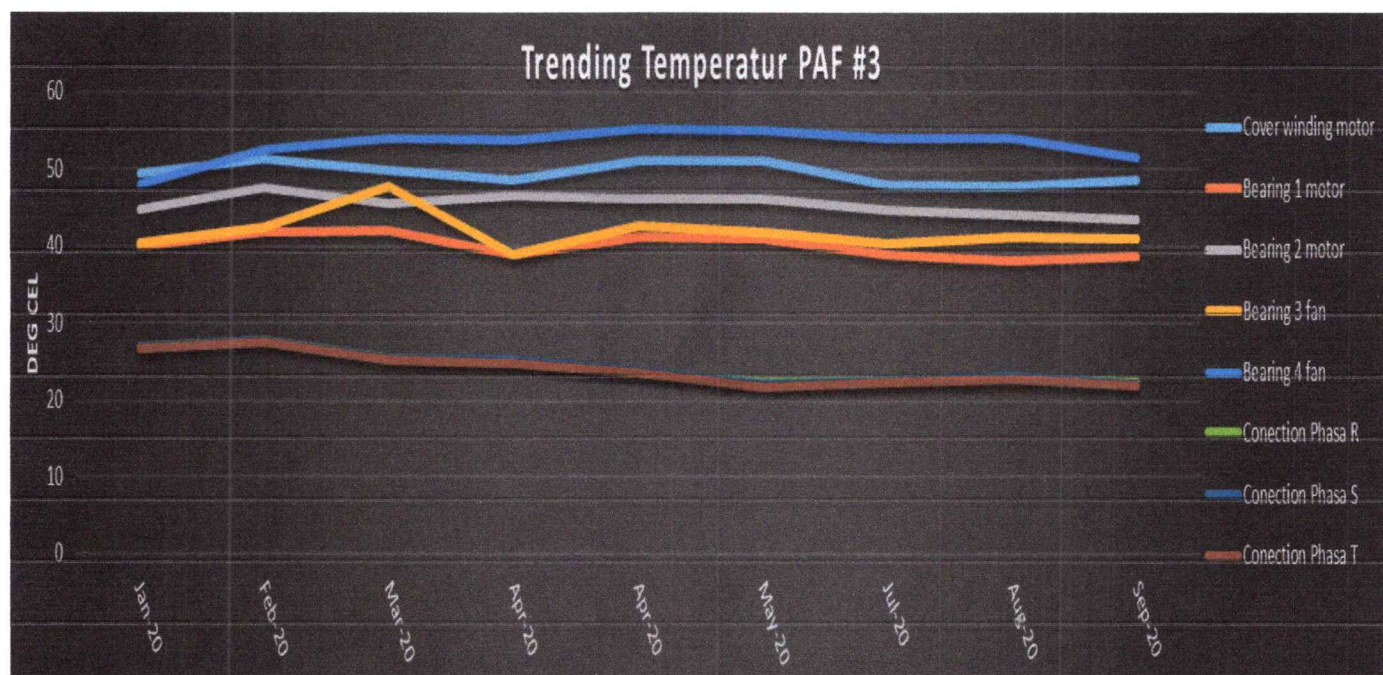
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III. Hasil Pengukuran IR Thermography

No. KKS	Description	Measurement Point	Upper Warning Limit Maximo	Upper Action Limit Maximo	Status per Point Measurement	Overall Status Equipment	T max Reference	Data Analysis	
								Date	(°C)
LI30HLB10AN010-001	EQUIPMENT-PRIMARY AIR FAN UNIT 3	Cover winding motor	71	81	Normal	Normal	60	16-Sep-20	48.6
		Bearing 1 motor	61	71	Normal		50	16-Sep-20	38.8
		Bearing 2 motor	61	71	Normal		50	16-Sep-20	43.4
		Bearing 3 fan	61	71	Normal		50	16-Sep-20	41.1
		Bearing 4 fan	61	71	Normal		50	16-Sep-20	51.5
		Conection Phasa R	41	51	Normal		30	16-Sep-20	22.4
		Conection Phasa S	41	51	Normal		30	16-Sep-20	22.3
		Conection Phasa T	41	51	Normal		30	16-Sep-20	22
		Current (Ampere)						16-Sep-20	34
		Damper (%)						16-Sep-20	4.6

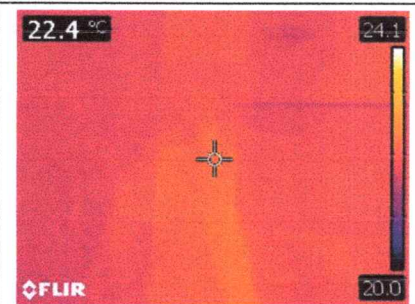
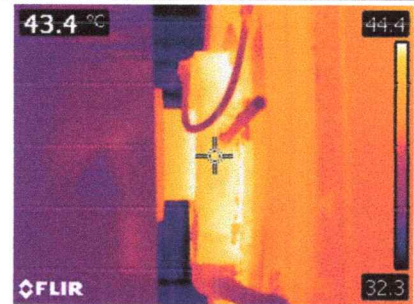
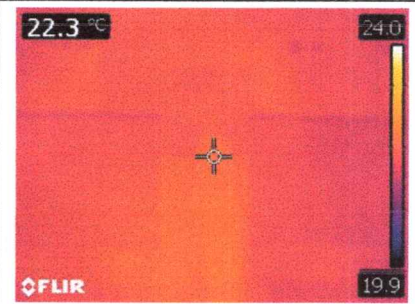
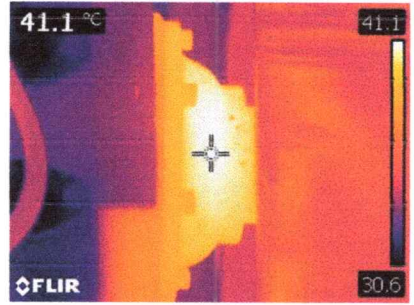


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CITRA IMAGING INFRARED THERMOGRAPHY	KETERANGAN	CITRA IMAGING INFRARED THERMOGRAPHY	KETERANGAN
	Pada body / cover winding PRIMARY AIR FAN 3 temp tertinggi 48.6 degC		Bearing 4 motor PRIMARY AIR FAN 3 temp tertinggi 51.5 degC
	Bearing 1 motor PRIMARY AIR FAN 3 temp tertinggi 38.8 degC		Phasa R PRIMARY AIR FAN 3 temp 22.4 degC
	Bearing 2 motor PRIMARY AIR FAN 3 temp tertinggi 43.4 degC		Phasa S PRIMARY AIR FAN 3 temp 22.3 degC
	Bearing 3 motor PRIMARY AIR FAN 3 temp tertinggi 41.1 degC		Phasa T PRIMARY AIR FAN 3 temp 22 degC

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IV. Pengukuran Tribologi :

Reference Oil	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46
Sample Date	1/27/2020	3/26/2020	4/22/2020	5/20/2020	6/19/2020	7/17/2020	9/12/2020
Machine Wear							
Total Fe, ppm	23	37	0	1	0	0	112
Fe >25, pml	29	6	9	2	1	0	41
Fe >38, pml	14	1	2	1	0	0	29
Fe >50, pml							
Fe >100, pml							
Wear Class							
MxD Ct >20, pml	22	9	12	13	0	0	478
MxD Ct >50, pml	1	0	0	1	0	0	14
MxD Ct >100, pml	1	0	0	1	0	0	0
MxD SI >20, pml	66	57	32	33	7	7	4089
MxD SI >50, pml	5	6	0	1	2	2	130
MxD SI >100, pml	1	0	0	1	0	0	6
MxD Ft >20, pml	179	89	56	57	8	7	9065
MxD Ft >50, pml	8	2	1	2	3	2	119
MxD Ft >100, pml	1	0	0	1	0	0	0
Contamination							
ISO >4	22	18	19	19	19	19	28
ISO >6	20	16	15	16	16	17	26
ISO >14	17	12	12	11	11	12	21
NAS 1638	11	8	9	9	9	9	12

!!!

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MxD Ox >20, pml	40	20	13	12	3	2	862
MxD Ox >50, pml	5	0	0	0	0	0	37
MxD Ox >100, pml	1	0	0	0	0	0	2
Cnts >4, pml	22835	36928	23915	21141	3564	4765	1657044
Cnts >6, pml	7817	5955	4638	3272	542	591	461739
Cnts >10, pml	1121	525	312	148	41	32	54177
Cnts >14, pml	352	194	98	74	18	21	15237
Cnts >18, pml	141	85	56	41	7	6	5085
Cnts >21, pml	73	38	21	16	6	6	1941
Cnts >26, pml	37	18	11	11	6	5	886
Cnts >32, pml	10	11	8	5	2	5	370
Cnts >38, pml	2	6	5	7	0	2	156
Cnts >56, pml	0.4	1.4	1.0	1.0	3.2	3.4	12.3
Cnts >70, pml	0.1	0.5	1.0	1.0	1.0	1.0	2.5
Water, ppm	312.0590	889.9302	619.3225	719.4901			
Chemistry							
Visc 40C, cSt	41.2	49.5	43.4	44.1	44.1	44.7	46.9
TAN, mg KOH/g	0.39	0.3	0.1	0.1			
IR Oxid., Abs/mm2	1.01	1.04	1.12	1.14			

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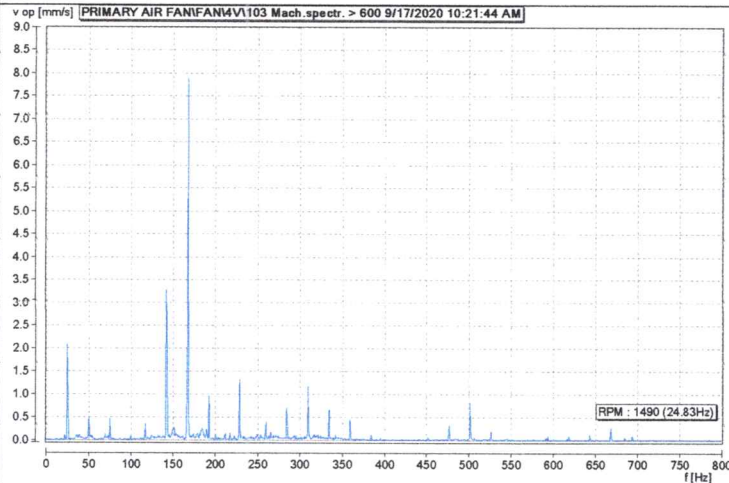
Reference Oil	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46	Pertamina - TURBO LUBE - 46
Sample Date	12/27/2019	1/27/2019	2/12/2020	3/26/2020	4/22/2020	5/25/2020	6/19/2020	7/17/2020	9/17/2020
Tapping Point	PAF	PAF	PAF	PAF	PAF	PAF	PAF	PAF	PAF
Elemental Analysis									
Machine Wear									
Aluminum, ppm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Antimony, ppm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Cadmium, ppm	0.00	0.00	0.00	0.00	0.00	0.00	0.28	0.23	0.28
Chromium, ppm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Copper, ppm	0.06	0.07	0.29	0.00	0.29	0.01	0.27	0.25	0.26
Iron, ppm	0.3	0.4	11.7	0.00	11.7	2.4	2.2	1.6	1.9
Lead, ppm	0.50	0.51	0.53	0.00	0.53	1.12	1.40	1.80	1.87
Manganese, ppm	2.52	2.59	2.43	9.90	2.43	1.27	1.32	1.21	1.63
Tin, ppm	0.00	0.01	0.00	0.00	0.00	0	0.11	0.13	0.17
Nickel, ppm	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00
Silver, ppm	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	1.84
Titanium, ppm	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Vanadium, ppm	0.00	0	0.07	0.00	0.07	0	0.00	0.00	0.00
Contamination									
Sodium, ppm	0.00	0.01	0.01	0.00	0.01	0.20	0.09	0.04	0.05
Silicon, ppm	2.08	3.08	4.48	0.00	4.48	7.77	0.43	1.23	1.72
Lithium, ppm	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00
Potassium, ppm	0.56	1.56	0.05	0	0.65	0.15	0.29	0.25	0.28
Chemistry									
Barium, ppm	0.20	0.21	0.25	0	0.25	0.15	0.12	0.17	0.23
Boron, ppm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Calcium, ppm	0.00	0.01	0.00	0.00	0.00	442	2.01	2.04	3.06
Magnesium, ppm	0.93	0.94	1.20	0.00	1.20	0.46	0.03	0.02	0.00
Molybdenum, ppm	0.00	0.01	0.00	0.00	0.00	1.01	1.21	1.06	0.00
Phosphorus, ppm	6.48	6.49	74.06	0.00	74.06	0.00	2.32	2.12	3.19
Zinc, ppm	1.62	1.63	71.63	0.00	71.63	1.03	0.42	0.23	0.28

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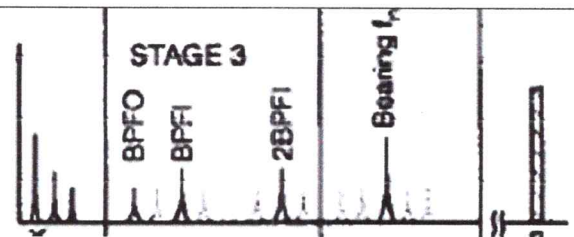
V. Analisa

a. Teknologi vibrasi

Syntom pengukuran



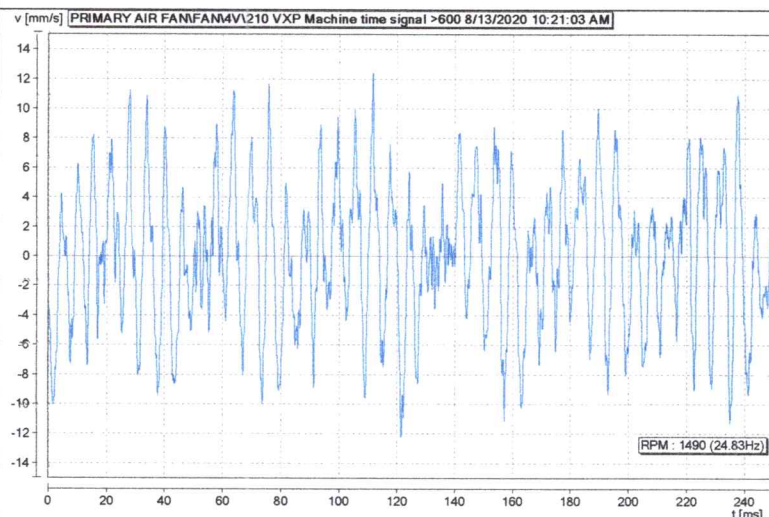
Syntom carlotte



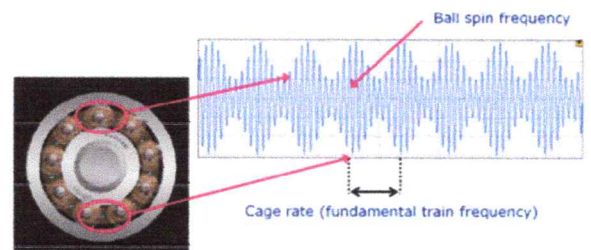
STAGE 3: Bearing defect frequencies and harmonics appear. When wear progresses, more defect frequency harmonics appear and number of sidebands grow, both around these and bearing component natural frequencies. Overall spike energy continues to increase (for example, from .5 to over 1 gSE). Wear is now usually visible and may extend throughout periphery of bearing, particularly when many well formed sidebands accompany bearing defect frequency harmonics. High frequency demodulated and enveloped spectra help confirm Stage III. *Replace bearings now! (independent of bearing defect frequency amplitudes in vibration spectra).*

Dari syntom pengukuran dibandingkan dengan pola charlotte mengindikasikan terjadi kerusakan bearing stage 3

Time Waveform Pengukuran



Time Waveform Analysis



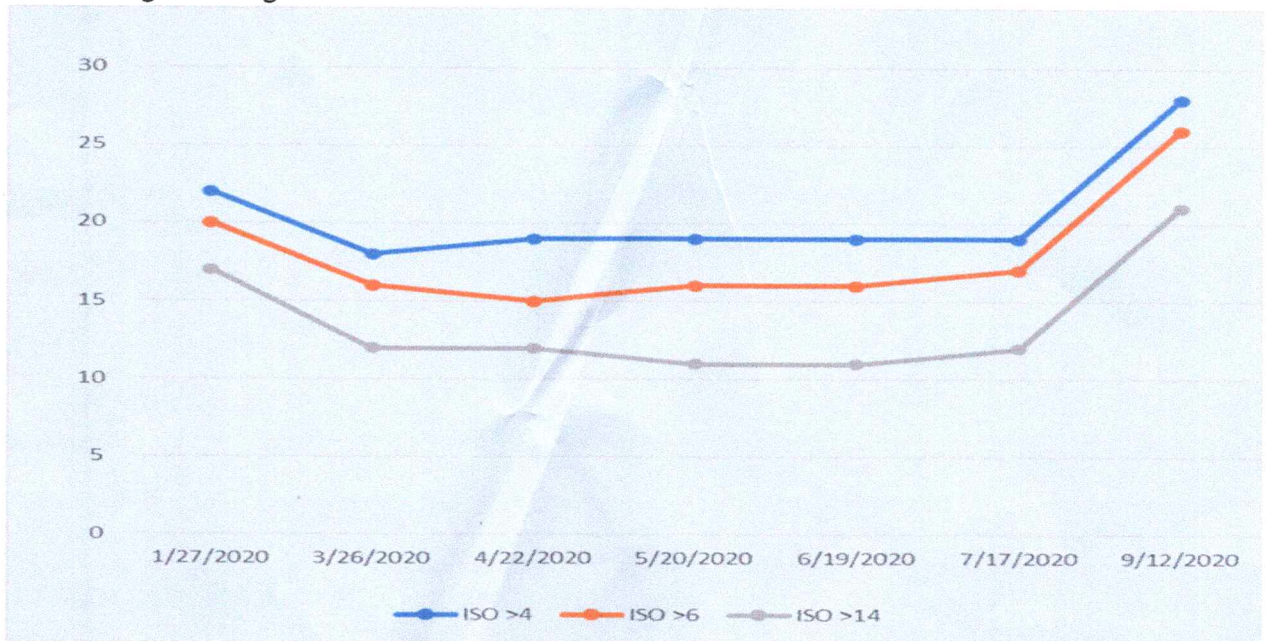
The result is that the amplitude of the vibration rises and falls periodically. The "carrier" frequency is the ball spin frequency, and it is modulated by the cage frequency (the rate at which the ball moves around the bearing). The spectrum has a peak at the "ball spin" frequency, and another peak at the "fundamental cage" or "FT" sidebands.

Dari Analysis time waveform pengukuran dibandingkan dengan pola timewave form reference mobius mengindikasikan terjadi kerusakan bearing

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b. Teknologi IR Thermography :
tidak ada kenaikan temperature pada PAF unit 3

c. Teknologi Tribologi :



(NAS 9) → (NAS 12)

1. Adanya kenaikan nilai partikel count dari 19/17/12 menjadi 28/26/21
2. Nilai Fe ada kenaikan dari 0 ppm menjadi 112 ppm
3. Nilai Manganese dari 1.21 ppm menjadi 1.63 ppm
4. Nilai Copper dari 0.25 ppm menjadi 3.63 ppm

IV. Kesimpulan :

Kemungkinan terjadi kerusakan Bearing 4 dan keausan pada shaft, dilihat dari spektrum vibrasi dan hasil pengujian oli.

Chemical Composition	Sumber kandungan
Manganese	Indikasi dari Shaft
Fe	Indikasi dari Bearing dan Shaft
Copper	Indikasi dari Bearing
Lead	Indikasi dari Bearing

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V. Rekomendasi :

Rekomendasi	Time	PIC
Ganti Bearing 1,2,3 dan 4	Rencanakan → ME	PEMELIHARAN
Ganti oli (drain filling)	segera	PEMELIHARAN
Penambahan Grease Bearing 1&2	Segera	PEMELIHARAN → Koordinasi
Monitoring Vibrasi	2 minggu sekali	ENGINEERING
Monitoring Tribologi	2 minggu sekali	ENGINEERING

HL-Pd

JERANJANG, 17 SEPTEMBER 2020

MENGETAHUI



THEODORUS TIO W.

DISUSUN OLEH



HERMAWAN